**QP CODE : 25900897**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) REGULAR/IMPROVEMENT EXAM, NOVEMBER 2025

Discipline Specific Core Course(DSC)- **MG1DSCMBG101- WONDERS OF THE MICROBIAL WORLD**

**PART A – MCQ (1 mark each)**

1. Edward Jenner
2. Staphylococcus
3. Flagellin
4. Joseph Lister
5. *Thiobacillus ferrooxidans*
6. Sarcinae
7. Mueller–Hinton agar
8. Capsule appears as colourless halos around stained cells
9. Barrier filter
10. Sulfonamides

**PART B – Fill in the Blanks**

11. $62.8^0$C/$63^0$C for 30 minutes
12. Leeuwenhoek/Louis Pasteur
13. Crystal violet
14. Cell membrane
15. Agar

**PART C – True or False**

16. True
17. False
18. False
19. True
20. False

**PART D – Very Short Answers**

21. Paul Ehrlich – discovered Salvarsan; "Father of Chemotherapy"
22. Life arises spontaneously from non-living matter
23. *Staphylococcus aureus*, *E. coli* (any two non–acid-fast bacteria)
24. Crystal violet, Methylene blue
25. Reflects excitation light, transmits emitted fluorescence to improve contrast
26. A thin film of microorganisms on a slide
27. Dormant, highly resistant bacterial structures
28. Fungi without a sexual reproductive stage
29. Diagram: TMV – rod-shaped, helical RNA virus.
30. Chlorine forms HOCl; oxidizes enzymes, proteins, and damages DNA
31. 121°C for 15 minutes at 15 psi
32. Controls the amount of light entering the condenser

**PART E – Short Answers (3 marks)**

33. Contributions of Louis Pasteur: Germ theory, pasteurization, vaccines, fermentation studies etc ( any six contributions, 0.5 marks for each)
34. Procedure of simple staining: smear → air dry → heat fix → apply stain → rinse → observe (3marks)
35. Resolution affected by: shorter wavelength, higher numerical aperture, better lens quality. (3marks)

36. Hanging drop vs wet mount: true motility vs Brownian motion (1.5marks)
    ; hanging drop lasts longer (1.5marks)

37. Bacterial cell structure: cell wall, membrane, cytoplasm, ribosomes, nucleoid, plasmids, flagella. (3marks)

38. Principles of sterilization: protein denaturation, oxidation, DNA damage, membrane disruption. (3marks)

**PART F – Long Answers (8 marks)**

39. MBC steps: MIC →(1mark) subculture clear tubes (3marks)→ incubate → lowest no-growth conc (4marks)= MBC
40. Intracytoplasmic features: nucleoid, ribosomes, plasmids, (4marks) inclusion bodies, endospores, cytoplasm. (4marks)